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MTH621

Real Analysis I

Mid- Term Preparation

Important Theorems For Short Questions & For Objective

Chapter # 01 The Real Number System

⇒ **Proper subset;**

A is proper subset of B if there exist at least one element of B which is not in A.

⇒ **Equal sets;**

"Two sets are said to be equal if they contain the same number of elements."

⇒ **Set of odd Number;**

$\{2k-1 : k \in N\}$

⇒ **Set of Even number;**

$\{2k : k \in N\}$

⇒ A set with only one member x_0 is Singleton set, denoted by $\{x_0\}$.

=> Prime Number are the building blocks of the positive Integers.

=> An equation with the Added provision that only Integer solution are sought is called Diaphantine.

i.e $a^n + b^n = c^n$, $n \in \mathbb{Z}$

=> **Mathematical Induction;**

let $P(n)$ be a Mathemtical statement, where $n \in \mathbb{N}$ (or $\mathbb{Z}^+$) if

① $P(1)$ is true

② $P(n)$ is true Implies $P(n+1)$ is true.

Examples;

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}$$

=> **An initial segment:**

An Initial segment I of $\mathbb{N}$ is a non-empty subset of $\mathbb{N}$ with the property that if $n \in I$ and $m \le n$ then $m \in I$.

② ⇒ Finite and Infinite set;

A set A is Finite if either A is empty or there exists $n \in N$ and a Bijective Mapping $c: I_n \to A$.

OR

A set S is said to be Finite if it is either empty or it has n elements for some $n \in N$.

⇒ Infinite :-

A set is if infinite if it is not Finite.

⇒ If $g: I_m \to I_n$ is an one to one (Injective) maping then $\boxed{m \le n}$

⇒ If A is a non-empty Finite set, there exist a unique $n \in N$ for which there exist a bijection $c: I_n \to A$.

⇒ The Number n is known as Cardinality of A and is written as $|A|$.

⇒ Proposition :-

Suppose (A) is Finite set, and that $F: A \to B$ is a bijective. Then B is finite. and $|B| = |A|$

Proof:- For if $C: I_{|A|} \to A$ is a bijective, then the mapping

$$f \circ C : I_{|A|} \to B$$ is the bijection.

⇒ **Field;**

A Field is a set F, together with two laws of Composition, addition $(+)$ and Multiplication $(\cdot)$ such that for all $a, b, c \in F$ Following properties holds;

* Commutative
* associative
* distributive
* Identity.
* Inverse.

⇒ **Dedekind Infinite set;**

A set A to be infinite if there is an injective map $j: A \to A$ which is not onto (surjective), such sets are called Dedekind Infinite.

⇒ **Example:-** Show that N is Dedekind infinite.

Sol; The mapping $f: N \to N$ defined by $f(n) = 2n$ is injective, and Not Surjective.

③ ⇒ The set of Natural numbers N is infinite set.

⇒ Countable; If it is Finite set. otherwise unCountable.

⇒ Not every set is Coutable, It was Cantor who First showed in 1873.

⇒ If A be a non-empty set. Then the Following are equivalent;

(a) A is Coutable.

(b) There exists a surjection $f: N \to A$.

(c) There exists an injection $g: A \to N$.

⇒ Theorem; The set $N \times N$ is Countable.

proof; The mapping $f: N \times N \to N$ by

$$f(k, l) = 2^k 3^l$$

$N \times N$ is countable.

⇒ The set of Rational Numbers is defined as;

$$Q = \left\{ \frac{m}{n} \; ; \; m, n \in \mathbb{Z}, \, n \neq 0 \right\}$$

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Example; Find solution of the equation $p = \sqrt{2}$ in the set of Rational Numbers if possible;

proof;- If $\sqrt{2}$ is Rational Number.

So, $\sqrt{2} = \frac{p}{q}$

Taking square on both sides;

$$2 = \frac{p^2}{q^2} \Rightarrow p^2 = 2q^2 \rightarrow (1)$$

p^2 should be of the form $4p'^2$

$$4p'^2 = 2q^2$$

$$2p'^2 = q^2$$

$$2p'^2 = 4q'^2$$

$$p'^2 = 2q'^2$$

$$p = 2p'$$

$$q = 2q'$$

$$\frac{p}{q} = \frac{2p'}{2q'}$$

Hence $\sqrt{2}$ is a irrational Number.

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⇒ Partial Order;

A Partial order is a relation R on a set S such that For all $a,b,c \in S$ we have the Following;

* Reflexivity (aRa)
* Anti symmetric (if aRb and $bRa \Rightarrow a=b$)
* Transitive (aRb, $bRc \Rightarrow aRc$)

⇒ lower Bound;

Any smaller number of set. if b is a lower bound of E, then so is any smaller number.

⇒ Infimum:-

If α is a lower bound of E, but no number greater than α. Then α is an infimum of E, and we write.

$$\alpha = \inf E$$

⇒ ~~Supremum~~; uper Bound:-

If b is an uper bound of E, then so, is any larger number.

⇒ Supremum;

if B is a uper bound of E but no number is Less than B is an upper Bound of E, then B is supremum of E.

$$B = \sup E.$$

=) If S is the set of negative numbers, then any nonnegative number is an uper bound of S and super $S = 0$

=) if S_1 is the set of negative Integers, then any number a such that $a \geq -1$ is an uper bound of S_1 and $\sup S_1 = -1$.

=) The Supermum and Infimum of a may or may not belong to that set.

=) <u>least uper bound property;</u>

An order set S is said to have the least-uper bound property if ~~at~~ the Following is true;

If $E \subset S$, E is not empty, and E is bounded above, then $\sup E$ exists in S.

=) <u>Theorem</u>; (The Completeness Axiom)

if a non-empty set of Real Numbers is bounded above, then it has a supremum, this is called Completeness.

=) Real Number system is a Complete ordered Field.

=) <u>Theorem</u> :- If a non-empty set S of Real number is bounded above, then $\sup S$ is the unique real Number β s.t

- $x \leq \beta$ for all x in S.
- If $\varepsilon > 0$, there is an x_0 in S such that $x_0 > \beta - \varepsilon$

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⇒ The Archimedean property of $\mathbb{R}$:-

Theorem: If ρ and ε are positive, then $n\varepsilon > \rho$ for some integer n, $n\varepsilon \le \rho$

⇒ Dense set in $\mathbb{R}$:-

A set D is said to be dense in the set of Real Numbers if every open Interval (a, b) Contains a number of D.

⇒ Theorem:

The Rational Numbers are dense in $\mathbb{R}$, that is, if a & b are Real numbers with $a<b$, there is a Rational number p/q such that $a < p/q < b$.

⇒ set of Rational Number is not complete.

Theorem: product of a Rational and irrational number is irrational.

proof:

let r is Rational and r' is irrational number; $r = p/q$

so we can write down as;

$rr' = \frac{m}{n}$ ⇒ $m \neq 0$, $q \neq 0$, $r \neq 0$

$r' = \frac{qm}{pn}$ it's Condratiction.

so r' is irrational numbers:-

Theorem;

Sum of Rational and irrational Number is irrational.

proof;

let r is Rational and r' is an irrational Number,

$$r + r' = \frac{m}{n}$$

Because r is rational number so,

$$r = p/q$$

$$r' = \frac{m}{n} - p/q \Rightarrow r' = \frac{m-p}{nq}$$

it is Contradiction.

so $r + r'$ is an irrational.

Theorem;

For every real $x > 0$ and every integer $n > 0$ there is one and only one positive real y such that $y^n = x$

⑥ Theorem;

if a and b are any two Real Numbers, then

$$|a+b| \leq |a|+|b|$$

=> The Extended Real Number System;

* A nonempty set S of Real numbers is unbounded above if it has no upper bound.

* A nonempty set S of Real numbers is unbounded below if it has no lower bound.

* unbounded above;

~~inf~~ ~~$S = +\infty$~~ $\sup S = \infty$

* unbounded below;

~~if~~ $\inf S = -\infty$

=> Mathematical Induction Examples:-

Example;

$$1+2+\text{----}+n = \frac{n(n+1)}{2}$$

$p_1 = p(1)$

$$1 = \frac{1(1+1)}{2} \Rightarrow 1 = \frac{1(2)}{2} \Rightarrow 1 = \frac{2}{2} = 1$$

$P_2 \Rightarrow$ put $n = n+1$

$$(1+2+\text{----}+n)+(n+1) = \frac{(n+1)(n+1+1)}{2}$$

$$= \frac{(n+1)(n+2)}{2}$$

Example;

For each non-negative Integer n, let x_n be a Real number and suppose that;

$$|x_{n+1} - x_n| \leq r\,|x_n - x_{n-1}|, \; n \geq 1 \rightarrow ①$$

Show that;

$$|x_n - x_{n-1}| \leq r^{n-1}|x_1 - x_0| \text{ if } n \geq 1 \rightarrow ②$$

proof

For $n = 1, 2, 3$ we define

$$|x_{1+1} - x_1| \leq r\,|x_1 - x_{1-1}| \text{ if } n \geq 1$$

$$|x_2 - x_1| \leq r\,|x_1 - x_0|,$$

$$|x_3 - x_2| \leq r\,|x_2 - x_1| \leq r^2|x_1 - x_0|$$

$$|x_4 - x_3| \leq r\,|x_3 - x_2| \leq r^3|x_1 - x_0|$$

Indective Step For $P(n+)$

put $n = n+1$ then we have;

$$|x_{n+1+1} - x_{n+1}| \leq r\,|x_{n+1} - x_{n+1-1}|$$

$$|x_{n+2} - x_{n+1}| \leq r\,|x_{n+1} - x_n| \Rightarrow \text{eq } Ⓐ$$

put in eq ② $n = n+1$

$$|x_{n+1} - x_{n+1-1}| \leq r^{n+1-1}|x_1 - x_0|$$

$$|x_{n+1} - x_n| \leq r^n|x_1 - x_0| \rightarrow Ⓑ$$

Comparing Ⓐ & Ⓑ

$$|x_{n+2} - x_{n+1}| \leq |x_{n+1} - x_n| \leq r^n|x_1 - x_0|$$

Hence proved.

(7) **Example:-**

prove the proposition;

$3n+16 > 0$, $n \geq -5$

solve) $n = -5, -4, -3 \ldots$

for $P(-5)$ ~~ny~~ $n = -5$

$3(-5) + 16 > 0$

$-15 + 16 > 0$

$1 > 0$

Basis step is true.

put $n = n+1$

$3_{n+1} + 16 > 0$

$3n + 16 + 3 > 0$

$3n + 19 > 0$

Indective step is also true for given proposition

Example:-

let P_n be the proposition that;

$n! - 3^n > 0$, $n \geq 7$

Sol:- (1) P_n is true when we put $n = 7$

(2) For ~~$P_{(n+1)}$~~ $P(n+1)$

$(n+1)! - 3^{n+1} > 0 \Rightarrow n!(n+1)! - 3^{n+1} > 0$

$3^n (n+1) - 3^{n+1} > 0$ $\quad\quad \because n! - 3^n > 0$

$3^n(n+1) - 3^n \cdot 3 > 0$ $\quad\quad n! = 3^n$

$3^n(n+1-3) > 0 \Rightarrow 3^n(n-2) > 0$

so indective step is also True.

=) **Generalization of union and Intersection;**

* The union and Intersection of Finitely many sets are written as;

$$\bigcup_{k=1}^{n} S_k \text{ and } \bigcap_{k=1}^{n} S_k.$$

* The union and intersection of an infinite sequence $\{S_k\}_{k=1}^{\infty}$ of sets written as,

$$\bigcup_{k=1}^{\infty} S_k \text{ and } \bigcap_{k=1}^{\infty} S_k.$$

* Thee open Interval (a, ∞) and ~~(a, ∞)~~ $(-\infty, b)$ are semi-infinite.

* If a, and b are finite, and $(-\infty, \infty)$ is then entire real line.

=) **ε - Neighborhood:-**

(If x_0 is a real number and $\varepsilon > 0$ then the open Interval $(x_0 - \varepsilon, x_0 + \varepsilon)$ is an ε- Neighborhood of x_0.)

=) **Deleted Neighborhood:-**

A Deleted neighborhood of a point x_0 is a set that contain every point of Some neighborhood of x_0 except for x_0 itself.

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$\Rightarrow$ The union of open set is open.

$\Rightarrow$ The Intersection of closed sets is Closed.

$\Rightarrow$ x_0 is a limit point of S if every deleted Neighborhood of x_0 contains a point of S.

$\Rightarrow$ x_0 is a boundary point of S if every Neighbourhood of x_0 contains at least one point in S and one not in S. Denoted by ∂S.

$\Rightarrow$ The closure of S, denoted by $\bar{S}$.

$$\bar{S} = S \cup \partial S$$

$\Rightarrow$ x_0 is an Isolated point of S if $x_0 \in S$ and there is a Neighborhood of x_0 that contain no other point of S.

$\Rightarrow$ x_0 is exterior to S if x_0 is in the interior of S^c. The collection of such points is the exterior of S.

$\Rightarrow$ Open Covering:-

A collection $\mathcal{H}$ of open sets is an open covering of a set S if every point in S is contained in a set H belonging to $\mathcal{H}$;

$$S \subset \cup \{H \in \mathcal{H} : H \in \mathcal{H}\}$$

⇒ Heine-Borel theorem;

If $\mathcal{H}$ is an open Covering of a closed and bounded Subset S of the Real number then S has an open Covering $\mathcal{H}$ Consisting of Finitely many open sets belonging to $\mathcal{H}$.

⇒ Compact set;

A Closed and bounded set of real numbers is Called Compact.

⇒ Theorem;

A set is closed iff no point of S^c is a limit point of S.

⇒ A set S is Closed iff S Contains all of its limit points.

⇒ (Bolzano Weierstrass Theorem)

"Every Bounded infinite set of Real numbers has at least one limit point."

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Chapter # 02

Sequences and Series

⇒ Sequences:-

"An infinite Sequence of real numbers is a real-valued function defined on a set of Integers $n : n \geq k$."

MCQs:

$$\{S_n\}_k^\infty = \{S_k, S_{k+1}, \text{---}\}$$

$$\{(-1)^n\}_0^\infty = \{1, -1, 1, \text{---}, (-1)^n, \text{---}\},$$

$$\left\{\frac{1}{n-2}\right\}_3^\infty = \left\{1, \frac{1}{3}, \frac{1}{3} \text{---} \frac{1}{n-2}, \text{---}\right\}$$

The real number S_n is the nth term of the sequence.

$\left\{\frac{1}{n-2}\right\}_3^\infty$ and $\left\{\frac{1}{n}\right\}_1^\infty$ are Identical.

⇒ Convergent Sequence:-

A Sequence $\{S_n\}$ Converge to a limit L, if for every $\varepsilon > 0$ there is an Integer N such that;

$$|S_n - L| < \varepsilon \quad \text{if} \quad n \geq N$$

S_n is Convergent and write;

$$\lim_{n \to \infty} S_n = L.$$

* A Sequence that does not converge is called diverges or is divergent.

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=> The limit of a Convergent sequence is unique.

=> A sequence $\{S_n\}$ is said to be divergent to ∞ if for any Real number a, $S_n > a$ for large n and written as:

$$\lim_{n \to \infty} S_n = \infty$$

Similary => $\lim_{n \to \infty} S_n = -\infty$ if for any Real number a, $S_n < a$ For large n.

Example:- The sequence $\{n/2 + \frac{1}{n}\}$ diverges to ∞, since, if a is any Real number, then

$$\frac{n}{2} + \frac{1}{2} > a \text{ if } n \geq 2a$$

$$\lim_{n \to \infty} \left(\frac{n}{2} + \frac{1}{n}\right) = \infty$$

(10) **Bounded Sequence:**

A sequence (s_n) is bounded above if there is a Real Number b such that

$$s_n \le b \text{ for all } n,$$

Bounded below if there is a real Number a such that;

$$s_n \ge a \text{ for all } n.$$

=> A Convergent sequence is bounded.

=> A sequence $\{s_n\}$ converges to s if and only if every neighborhood of s contain s_n for all but Finitely many n.

=> Monotonic sequences;

"A sequence that is either nonincreasing or Nondecreasing is called Monotonic Sequences."

* A sequence $\{s_n\}$ is nondecreasing if $s_n \ge s_{n-1}$ for all n.
* A sequence $\{s_n\}$ is non-increasing if $s_n \le s_{n-1}$ for all n.

Theorem:-

* If $\{s_n\}$ is non-decreasing the $\lim_{n\to\infty} s_n = \sup\{s_n\}$

* If $\{s_n\}$ is non-Increasing, then $\lim_{n\to\infty} s_n = \inf\{s_n\}$.

$\Rightarrow$ if $s_n > s_{n-1}$ then $\{s_n\}$ is increasing.

$\Rightarrow$ if $s_n < s_{n-1}$ then $\{s_n\}$ is decreasing.

Theorem;

If $p > 0$ then

$$\lim_{n\to\infty} \frac{1}{n^p} = 0$$

Inequality From Binomial theorem;

$$1 + nx \le (1+x)^n, \quad x > 0$$

Theorem; Show that;

$$\lim_{n\to\infty} \sqrt[n]{n} = 1$$

proof;

Take $x_n = \sqrt[n]{n} - 1$

Then $x_n \ge 0$ and by the binomial theorem;

$$\cancel{n}(1 + x_n)^n \ge \frac{n(n-1)}{2} x_n^2$$

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$$x_n^2 \le \frac{2}{n-1}$$

$$0 \le x_n \le \sqrt{\frac{2}{n-1}}, \quad n \ge 2$$

if we take limit;

$$\lim_{n\to\infty} \sqrt{\frac{2}{n-1}} = 0$$

so, $0 \le x_n \le 0$

so $\lim_{n\to\infty} x_n = 0$.

Theorem;

if $p>0$ and α is real, the

$$\lim_{n\to\infty} \frac{n^\alpha}{(1+p)^n} = 0$$

proof; let k be an INteger such that

$k > \alpha$, $k > 0$ $\quad$ $\because n(n-1) \cdots n-(k+1) = \frac{n!}{2^k}$

for $n > 2k$.

$$(1+p)^n > \frac{n(n-1)\cdots n-(k+1)}{k!} p^k > \frac{n!\, p^k}{2^k k!}$$

$$(1+p^n) > \frac{n!\, p^k}{2^k k!} \Rightarrow \text{By using Reciprocal;}$$

$$\frac{1}{(1+p)^n} > 2^k k! / n^k p^k$$

$$\frac{1}{(1+p)^n} < \frac{n^{-k} 2^k k!}{p^k}$$

$$\frac{1 \times n^\alpha}{(1+p)^n} < \frac{n^{\alpha - k} \; 2^k k!}{p^k}$$

Hence,

$$0 < \frac{n^\alpha}{(1+p)^n} < \frac{2^k k!}{p^k} n^{\alpha-k}, \; n > 2k$$

Since $\alpha - k < 0$, $n^{\alpha-k} \to 0$ by

$$\lim_{n\to\infty} \frac{1}{n^p} = 0$$

Hence proved,

$$\lim_{n\to\infty} \frac{n^\alpha}{(1+p)^n} = 0$$

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Theorem;

Let $\lim_{n \to \infty} S_n = s$ and $\lim_{n \to \infty} t_n = t$ where s and t are finite.

then $\lim_{n \to \infty} (cs)_n = cs$

if c is constant;

$$\lim_{n \to \infty} (s_n + t_n) = s + t$$

$$\lim_{n \to \infty} (s_n - t_n) = s - t$$

Example:- Find the limit of the sequence

$$S_n = \frac{1}{n} \sin \frac{n\pi}{4} + \frac{2(1+3/n)}{1 + 1/n}$$

Solve; we apply the Applicable parts of Theorem;

$$\lim_{n \to \infty} s_n = \lim_{n \to \infty} \frac{1}{n} \frac{\sin n\pi}{4} + \frac{2\left[\left(\lim_{n \to \infty} 1 + 3 \lim_{n \to \infty} \left(\frac{1}{n}\right)\right)\right]}{\lim_{n \to \infty} 1 + \lim_{n \to \infty} \left(\frac{1}{n}\right)}$$

$$= 0 + \frac{2(1 + 3 \cdot 0)}{1 + 0} = 2$$

Example:-

Find the limit of the Subsequence

$$S_n = \lim_{n\to\infty} \frac{(n/2)+(\log n)}{3n+4\sqrt{n}}$$

$$= \lim_{n\to\infty} \frac{1/2 + (\log n)/n}{3+4n^{-1/2}}$$

$$= \frac{\lim\limits_{n\to\infty} 1/2 + \lim\limits_{n\to\infty} (\log n)/n}{\lim\limits_{n\to\infty} 3 + 4\lim\limits_{n\to\infty} n^{-1/2}}$$

$$= \frac{1/2+0}{3+0}$$

$$= 1/6$$

Subsequence:-

A sequence $\{t_k\}$ is a subsequence of a sequence $\{S_n\}$ if;

$$t_k = S_{n_k} \qquad k \geq 0$$

where $\{n_k\}$ is an increasing infinite sequence of integers in the domain of $\{S_n\}$.

we denote the Subsequence $\{t_k\}$ by $\{S_{n_k}\}$

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Example:

If $\{S_n\} = \{\frac{1}{n}\} = \{1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots\}$

let:

$n_k = 2k$

$\{S_{2k}\} = \{\frac{1}{2k}\} = \{\frac{1}{2}, \frac{1}{4}, \dots, \frac{1}{2k}, \dots\}$

$n_k = 2k+1$

$\{S_{2k+1}\} = \{\frac{1}{2k+1}\} = \{1, \frac{1}{3}, \frac{1}{5}, \dots, \frac{1}{2k+1}\}$

Theorem:

"Every subsequence of convergent sequence is convergent."

Theorem:

"If a sequence is Monotonic & has convergent subsequence then the sequence is convergent."

$$\lim_{k \to \infty} S_{n_k} = S,$$

then $$\lim_{n \to \infty} S_n = S.$$

limit point:

Theorem:

A point $\bar{x}$ is a limit point of a set S iff there is a sequence $\{x_n\}$ of point in S such that $x_n \neq \bar{x}$ for $n \geq 1$ and

$$\lim_{n \to \infty} x_n = \bar{x}.$$

Theorem:-

* if $\{x_n\}$ is bounded, then $\{x_n\}$ has a convergent subsequence.
* if $\{x_n\}$ is unbounded above, then $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ such that;

$$\lim_{k \to \infty} x_{n_k} = \infty$$

* if $\{x_n\}$ is unbounded below, then $\{x_n\}$ has a subsequence $\{x_{n_k}\}$ such that;

$$\lim_{k \to \infty} x_{n_k} = -\infty$$

(14) **limit Superior and limit Inferior;**

Theorem:-

* If $\{S_n\}$ is Bounded above and does not diverge to $-\infty$, then there is a unique Real Number $\bar{S}$, Such that, if $\varepsilon > 0$

$$S_n < \bar{S} + \varepsilon \quad \text{For large } n$$

and

$$S_n > \bar{S} - \varepsilon \quad \text{for infinitely many } n.$$

* If $\{S_n\}$ is bounded below and does not diverge to ∞, then there is a unique Real Number $\underline{S}$ Such that, if $\varepsilon > 0$.

$$S_n > \underline{S} - \varepsilon \quad \text{for lage } n$$

$$S_n \leq \underline{S} + \varepsilon \quad \text{for infinitely many } n.$$

limit superior and limit Inferior of $\{S_n\}$ is denoted by;

$$\bar{S} = \limsup_{n\to\infty} S_n \quad , \quad \underline{S} = \liminf_{n\to\infty} S_n$$

$\Rightarrow$ $\limsup_{n\to\infty} s_n = \infty \Rightarrow$ if $\{s_n\}$ is Not bounded above

$\Rightarrow$ $\limsup_{n\to\infty} s_n = -\infty \Rightarrow$ if $\lim_{n\to\infty} s_n = -\infty$

$\Rightarrow$ $\liminf_{n\to\infty} s_n = -\infty \Rightarrow$ if $\{s_n\}$ is not bounded ~~above~~ below

$\Rightarrow$ $\liminf_{n\to\infty} s_n = \infty \Rightarrow$ if $\lim_{n\to\infty} s_n = \infty$

Theorem;

Every sequence $\{s_n\}$ of Real Numbers has a unique limit Superior, $\overline{s}$ and limit inferior, $\underline{s}$, in the extended Reals,

$$\underline{s} \le \overline{s}.$$

$\Rightarrow$ **Cauchy Sequence;**

A sequence $\{s_n\}$ of Real numbers is called a cauchy sequence if for every $\varepsilon > 0$, there is an Integer N that; is

$$|s_n - s_m| < \varepsilon \text{ if } m, n \ge N.$$

$\Rightarrow$ **Theorem;**

if $\{s_n\}$ is a Cauchy sequence of Real Numbers, then $\{s_n\}$ is bounded.

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Theorem; A sequence $\{S_n\}$ of Real Numbers Converges iff, for every $\varepsilon > 0$ there is an Integer N such that;

$$|S_n - S_m| < \varepsilon \quad \text{if} \quad m, n \geq N$$

Series

Topic#60

=> **Oscillatory Series;**

A divergent infinite series that does not diverge to $\pm\infty$ is said to oscillate, or the oscillatory.

=> **Theorem;** The sum of a Convergent series is unique.

$$\Rightarrow \sum_{n=k}^{\infty} (ca_n) = CA$$

if c is Constant;

$$\Rightarrow \sum_{n=k}^{\infty} (a_n + b_n) = A + B$$

$$\Rightarrow \sum_{n=k}^{\infty} (a_n - b_n) = A - B$$

=> The Series $\sum_{k}^{\infty} a_n$ Converges if $(-1)^n a_n > 0$.

, $|a_{n+1}| < |a_n|$, and $\lim_{n \to \infty} a_n = 0$

=> Theorem:

A Series $\sum a_n$ Converges iff for every $\varepsilon > 0$ there is an Integer N such that:

$$|a_n + a_{n+1} ---- a_m| < \varepsilon \text{ if}$$

$$m \geq n \geq N.$$

=> $\sum a_n$ Converges iff $\{A_n\}$ converges.

Theorem;

Show that the series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent

$S_1 = 1$, $S_2 = 1 + \frac{1}{2}$, $S_3 = 1 + \frac{1}{2} + \frac{1}{3}$, $S_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$, ---

Form strictly increasing sequence.

$$S_1 < S_2 < S_3 < S_4 < --- < S_n < ---$$

$S_{2^1} = 1 + \frac{1}{2} > \frac{1}{2} + \frac{1}{2}$

$S_4 = 2^2 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \geq S_2 + \frac{1}{3} + \frac{1}{4} \geq S_2 + \frac{1}{4} + \frac{1}{4}$

$$S_8 = 2^3 = S_4 + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > S_4 + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} > \frac{4}{8}$$

$$\sim S_{2^n} > \frac{n+1}{2}$$

If M is any Constant, we Can Find a positive Integer n such that;

$$\frac{n+1}{2} > M.$$

for n we have

$$S_{2^n} > \frac{n+1}{2} > M$$

Example:

If $\sum a_n$ Converges, then $\lim\limits_{n \to \infty} a_n = 0$

Proof:, If $\sum a_n$ Converges, then for each $\varepsilon > 0$ there is an Integer K such that;

$$\left| \sum_{n \geq k}^{\infty} a_n \right| < \varepsilon \quad \text{if} \quad k \geq K;$$

that is

$$\lim_{k \to \infty} \sum_{n \geq k}^{\infty} a_n = 0$$

$\lim\limits_{n \to \infty} = 0$ Necessary Condition not sufficient.

⇒ Series of Non-Negative;

"The Series $\sum a_n$ is said to be Series of Non-Negative terms if $a_n \geq 0$ for $n \geq k$."

"The series of Non-Negative term either Converges to a Finite limit or diverges to ∞."

The Comparsion Test

Theorem

Suppose that

$$0 \leq a_n \leq b_n, \quad n \geq k,$$

Then;

* $\sum a_n < \infty$ if $\sum b_n < \infty$
* $\sum b_n = \infty$ if $\sum a_n = \infty$

Proof ;-

If ;

$A_n = a_k + a_{k+1} + \text{---} + a_n$ and

$B_n = b_k + b_{k+1} + \text{---} + b_n$, $\underline{n \geq k}$

then, we have;

$$A_n \leq B_n$$

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$\Rightarrow$ If $\sum b_n < \infty$, then $\{B_n\}$ is Bounded above this implies that $\{A_n\}$ is also, therefore $\sum a_n < \infty$.

$\Rightarrow$ If $\sum a_n = \infty$ then $\{A_n\}$ is unbounded above implies that $\{B_n\}$ is also, therefore, $\sum b_n = \infty$

Example :-

Discuss the Convergence of the Series $\sum \frac{r^n}{n}$.

Since :,

$$\frac{r^n}{n} < r^n \quad , \quad n \geq 1$$

$\sum r^n < \infty$ ~~the~~ if $0 < r < 1$, the Series $\sum r^n/n$ converges if $0 < r < 1$.

The Series is inconclusive (mean we cant say anything about convergent or divergent of series) if $r > 1$

If $r < 0$ Comparsion test does not apply. Since the Series have infinitely many Negative terms.

Theorem;

Suppose $a_1 \geq a_2 \geq \cdots \geq 0$. Then the Series $\sum_{n=1}^{\infty} a_n$ Converges iff the Series $\sum_{k=0}^{\infty} 2^k a_{2^k}$ Converges.

Theorem;

The Series $\sum \frac{1}{n^p}$ Converges if $p>1$ and diverges if $p \leq 1$

Theorem;

if $p > 1$

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p} \quad \text{Converges.}$$

if $p<1$, the Series diverges.

Proof:-

The Monotonicity of the logarithmic Function implies $\{\log n\}$ increases. Hence $\{1/n\log n\}$ decreases.

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$$

By using theorem;

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∴ Suppose $a_1 \geq a_2 \geq \cdots \geq 0$ then the Series $\sum_{n=1}^{\infty} a_n$ converges iff the Series $\sum_{k=0}^{\infty} 2^k a_{2^k}$ converges.

$$a_n = \frac{1}{n(\log n)} \Rightarrow a_{2^k} = \frac{1}{2^k(\log 2^k)^p}$$

$\Downarrow$

∴ By using log property;

$\Rightarrow \log m^n$

$n \log m$

So the Series;

$$a_{2^k} = \frac{1}{2^k (k\log 2)^p}$$

$$\sum_{k=1}^{\infty} \frac{2^k}{2^k(k\log 2)^p} = \frac{1}{(\log 2)^p} \sum_{k=1}^{\infty} \frac{1}{k^p}$$

So By using the result, of Harmonic Series.

our given Series is Convergent if $P > 1$ and divergent $P < 1$.

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(The Number e) Imp long

The number e is defined by infinite Series and is given by;

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}$$

where;

$n! = 1.2.3 --- n$ if $n \geq 1$ and $0! = 1$

(Approximation of Number 'e') Imp long

let the sequence of partial sum!

$$S_n = 1 + 1 + \frac{1}{1.2} + \frac{1}{1.2.3} + --- + \frac{1}{1.2.3--n}$$

if we subtract (e) from this partial sum

All term upto (n) terms will be cancel out and then we have;

$$e - S_n = \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \frac{1}{(n+3)!} + ---$$

$$< \frac{1}{(n+1)!}\left(1 + \frac{1}{n+1} + \frac{1}{(n+1)^2} + ---\right) = \frac{1}{n!n}$$

so, we can conclude that;

$$0 < e - S_n < \frac{1}{n!n}.$$

$\Rightarrow$ The number e is an irrational number.

Theorem;

Suppose that $a_n \geq 0$ and $b_n > 0$ for $n \geq k$. Then,

* $\sum a_n < \infty$ if $\sum b_n < \infty$ and $\limsup_{n \to \infty} a_n/b_n < \infty$

* $\sum a_n = \infty$ if $\sum b_n = \infty$ and $\liminf_{n \to \infty} a_n/b_n > 0$

Example;

$$\sum b_n = \sum \frac{1}{n^{p+q}} \quad \text{and} \quad \sum a_n = \sum \frac{2 + \sin n\pi/6}{(n+1)^p (n-1)^q}$$

Sol,

$$\frac{a_n}{b_n} = \frac{2 + \sin n\pi/6}{(1 + 1/n)^p (1 - 1/n)^q}$$

$$\lim_{n \to \infty} \frac{a_n}{b_n} = 3, \quad \lim_{n \to \infty} \frac{a_n}{b_n} = 1.$$

Since $\sum b_n < \infty$ iff $p + q > 1$, the same is true of $\sum a_n$.

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⇒ The Ratio Test:-

Theorem: Suppose that $a_n > 0$, $b_n > 0$ and

$$\frac{a_{n+1}}{a_n} \leq \frac{b_{n+1}}{b_1}$$

we can also write it as:

$$\frac{a_{n+1}}{b_{n+1}} \leq \frac{a_n}{b_n} \Rightarrow \left\{\frac{a_n}{b_n}\right. \text{ is Non-increasing.}$$

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Then:

* $\sum a_n < \infty$ if $\sum b_n < \infty$
* $\sum b_n = \infty$ if $\sum a_n = \infty$

⇒ Suppose that $a_n > 0$ $(n \geq k)$ and

$$\lim_{n \to \infty} \frac{a_{n+1}}{a_n} = L$$

Then

$$\sum a_n < \infty \text{ if } L < 1.$$

$$\sum a_n = \infty \text{ if } L > 1.$$